



A high-order multi-time-step scheme for bond-based peridynamics

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ARTICLE INFO

Keywords:

Bond-based peridynamics
Multi-time-step
Higher-order method
Crack propagation

ABSTRACT

A high-order multi-time-step scheme (MTS) for the bond-based peridynamic (PD) model, an extension of classical continuous mechanics widely used for analyzing discontinuous problems like cracks, is proposed. The MTS scheme discretizes the spatial domain with a meshfree method and advances in time with a high-order Runge–Kutta method. To effectively handle discontinuities (cracks) that appear in a local subdomain in the solution, the scheme employs the Taylor expansion and Lagrange interpolation polynomials with a finer time step size, that is, coarse and fine time step sizes for smooth and discontinuous subdomains, respectively, to achieve accurate and efficient simulations. By eliminating unnecessary fine-scale resolution imposed on the entire domain, the MTS scheme outperforms the standard STS scheme for PD by significantly reducing computational costs, particularly for problems with discontinuous solutions, as demonstrated by comprehensive theoretical analysis and numerical experiments.

1. Introduction

In classical continuum mechanics, motion equations are expressed as partial differential equations involving spatial derivatives of the displacement field. Nevertheless, when confronted with discontinuous problems such as damage and fracture, the spatial derivatives of displacement components demonstrate nonphysical singularities in the discontinuous regions. Numerous advanced numerical methods have been developed to address this issue. The finite element method (FEM) stands out as a widely-used approach for predicting cracking behavior in various solids [1,2]. However, FEM relies on re-meshing technology to capture crack propagation paths, and extrinsic criteria must be incorporated into the finite element model to determine parameters such as crack propagation rate, branching, and coalescence [3]. Various enhanced numerical methods, such as the extended finite element method (XFEM) [4,5], have been proposed in response to these challenges. XFEM addresses the limitations of re-meshing, yet it still encounters challenges in accurately simulating the propagation and interaction of multiple cracks within complex patterns. The phase-field method [6,7] recently has gained popularity among researchers for studying crack mechanisms in various materials. This method's capability to represent cracks with a diffusive region surpasses other treatments. However, the phase-field method is more suitable for 2D problems than 3D ones and demands extensive mesh refinement along crack paths, resulting in substantial computational costs [8].

On the other hand, peridynamics has garnered increasing attention in recent years as an efficient tool for modeling cracking and failure processes. In this class of methods, the finite element method solves integral motion equations instead of differential equations, simplifying the consideration of discontinuities associated with the cracking process. Initially, peridynamic models were

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<https://doi.org/10.1016/j.cam.2024.115968>

Received 29 July 2023; Received in revised form 27 March 2024

Available online 26 April 2024

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bond-based [9], requiring materials to have fixed Poisson's ratios. Later, state-based models [10] were introduced, eliminating the limitation of Poisson's ratio. The versatility of peridynamic models has been demonstrated in various practical applications, such as deformation analysis of composite materials [11–13], investigation of corrosion phenomena [14–18], prediction of damage [19–23], and simulation of crack propagation in diverse materials [24–28].

Various spatial discretization approaches have been employed in peridynamics, including meshfree, collocation, finite element, and finite difference methods [29–33]. Researchers have also explored algorithms based on fast Fourier transform [34,35] and coupling methods [36–41] to alleviate the computational burden. For temporal discretization, adaptive dynamic relaxation (ADR) and velocity Verlet (VV) methods are commonly used methods for solving quasi-static and dynamic problems, respectively, providing accurate and efficient solutions in peridynamics.

However, when dealing with large and complex nonlocal problems, the computational burden of the aforementioned numerical methods can be significant, especially when using a small global single-time-step (STS) to achieve temporal accuracy. As a solution, multi-time-step (MTS) schemes have gained popularity in recent years. One notable example is the second-order MTS scheme, which builds upon the second-order strong stability preserving Runge–Kutta (SSP-RK) method for shallow-water equations [42]. Another MTS scheme based on the second-order velocity Verlet (VV) algorithms improves computational efficiency by utilizing variable time step sizes for different material components in solving PD [43]. Those are second-order schemes in time. In [44], a high-order STS scheme for reformulations of both bond-based and state-based PD modeling is proposed, showing that a high-order STS scheme not only improves the accuracy and the efficiency for benchmark smooth problems but also exhibits a promising behavior for some non-smooth models such as PD.

This work presents a high-order MTS scheme for PD. In this study, we locally adapt the fine time step size near the crack, reducing the overall computational costs, and ensuring accuracy. By incorporating MTS schemes in time, the proposed method achieves comparable accuracy to STS schemes, such as RK4, in a lesser wall-clock time by reducing the overall computational cost of PD simulations. We validate the feasibility and performance of the proposed high-order MTS scheme through extensive numerical examples, demonstrating its efficacy in solving PD problems.

The article is organized as follows: Section 2 reviews the PD and the temporal discretization schemes. Section 3 analyzes the MTS scheme based on the third- and fourth-order Runge–Kutta for the PD and provides proof of convergence order. Section 4 offers numerical examples demonstrating the accuracy and convergence of the MTS scheme. Finally, Section 5 provides the conclusion.

2. Fundamentals of bond-based peridynamics and its discretization

This section provides a brief overview of the linearized and nonlinearized versions of the bond-based PD, which are the main focuses of this study. Additionally, it briefly outlines the spatial discretization used to solve these models. The equation of motion for the bond-based PD is given below:

$$\rho \frac{d^2 \mathbf{u}}{dt^2}(\mathbf{x}_i, t) = \int_{H_{\mathbf{x}}} \mathbf{p}(\mathbf{u}(\mathbf{x}', t) - \mathbf{u}(\mathbf{x}, t), \mathbf{x}' - \mathbf{x}) dV_{\mathbf{x}'} + \mathbf{b}(\mathbf{x}, t), \quad (1)$$

where ρ is the mass density, $H_{\mathbf{x}}$ is the horizon, which is usually taken as a disk or ball of radius δ . $\mathbf{u}(\mathbf{x}, t)$ and $\mathbf{b}(\mathbf{x}, t)$ are the displacement vector and body force density fields, respectively. Let denote $\xi = \mathbf{x}' - \mathbf{x}$ and $\eta = \mathbf{u}(\mathbf{x}', t) - \mathbf{u}(\mathbf{x}, t)$, then the pairwise (linear and nonlinear) force function $\mathbf{p}$ exerted by a particle $\mathbf{x}$ on another particle $\mathbf{x}'$ can be expressed as follows [45]:

- linearized form

$$\mathbf{p} = \alpha \frac{\xi \otimes \xi}{\|\xi\|^3} \eta, \quad (2)$$

- nonlinearized form

$$\mathbf{p} = \alpha \frac{\|\xi + \eta\| - \|\xi\|}{\|\xi\|} \frac{\xi + \eta}{\|\xi + \eta\|}. \quad (3)$$

Here, α represents a scalar parameter introduced to ensure equivalence in energy between the PD and the classical elastic model, as outlined below:

$$\alpha = \begin{cases} \frac{9E}{\pi\delta^3 d}, & 2D, \\ \frac{12E}{\pi\delta^4}, & 3D, \end{cases} \quad (4)$$

where E is the elastic modulus, ν is the Poisson's ratio, and d is the thickness of the plate.

For discretizing (1) at a material point $\mathbf{x}_i$ for the linear bond-based PD, a meshfree method is employed for its simplicity in which the semi-discrete system can be rewritten as

$$\rho \frac{d^2 \mathbf{u}_i}{dt^2} = \sum_{\mathbf{x}_j \in H_{\mathbf{x}_i}} \alpha \frac{(\mathbf{x}_j - \mathbf{x}_i) \otimes (\mathbf{x}_j - \mathbf{x}_i)}{\|\mathbf{x}_j - \mathbf{x}_i\|^3} (\mathbf{u}_j - \mathbf{u}_i) V_j + \mathbf{b}(\mathbf{x}_i, t), \quad (5)$$

where $\mathbf{u}_i = \mathbf{u}(\mathbf{x}_i, t)$ and V_j is the volume of the material point $\mathbf{x}_j$.

Table 1
Butcher Tableau for the explicit r -order r -stages RK method.

0				
c_2	a_{21}			
c_3	a_{31}	a_{32}		
$\vdots$	$\vdots$		$\ddots$	
c_r	a_{r1}	a_{r2}	$\dots$	$a_{r,r-1}$
	b_1	b_2	$\dots$	b_r

Table 2
Butcher Tableau for RK3.

0			
2/3	2/3		
2/3	0	2/3	
	1/4	3/8	3/8

Similarly, for the nonlinear bond-based PD, (1) can be discretized as

$$\rho \frac{d^2 \mathbf{u}_i}{dt^2} = \sum_{\mathbf{x}_j \in \mathcal{H}_{\mathbf{x}_i}} \alpha \frac{\|\mathbf{x}_j + \mathbf{u}_j - \mathbf{x}_i + \mathbf{u}_i\| - \|\mathbf{x}_j - \mathbf{x}_i\|}{\|\mathbf{x}_j - \mathbf{x}_i\|} \frac{\mathbf{x}_j + \mathbf{u}_j - \mathbf{x}_i + \mathbf{u}_i}{\|\mathbf{x}_j + \mathbf{u}_j - \mathbf{x}_i + \mathbf{u}_i\|} V_j + \mathbf{b}(\mathbf{x}_i, t). \quad (6)$$

By denoting $\mathbf{Q} = \{\mathbf{U}_j(t), \mathbf{x}_j \in \mathcal{H}_{\mathbf{x}_i}\}$, where $\mathbf{U}_i(t) = [\mathbf{u}(\mathbf{x}_i, t), \mathbf{v}(\mathbf{x}_i, t)]^T$, we can express the original semi-discrete system compactly as:

$$\frac{d\mathbf{U}_i}{dt} = \mathcal{L}_i(\mathbf{Q}), \quad (7)$$

where $\mathcal{L}$ is the spatial operator such that

$$\mathcal{L}_i(\mathbf{Q}) = \begin{bmatrix} \left(\sum_{\mathbf{x}_j \in \mathcal{H}_{\mathbf{x}_i}} \mathbf{p}(\mathbf{u}_j - \mathbf{u}_i, \mathbf{x}_j - \mathbf{x}_i) V_j + \mathbf{b}(\mathbf{x}_i, t) \right) / \rho \\ \mathbf{v}(\mathbf{x}_i, t) \end{bmatrix}, \quad (8)$$

in which $\mathbf{v} = \dot{\mathbf{u}}$.

2.1. Temporal discretization schemes based on the Runge–Kutta methods

This section presents a high-order temporal discretization scheme, without loss of generality and for clarity, using a uniformly-sized partition of the time interval $[0, T]$ into N subintervals:

$$0 = t_0 \leq t_1 \leq \dots \leq t_{N-1} \leq t_N = T, \quad \Delta t = t_{n+1} - t_n, \quad 0 \leq n \leq N. \quad (9)$$

To discretize the system described by Eqs. (5) and (6) over time, we can use high-order Runge–Kutta (RK) methods that are commonly used to solve systems of ordinary differential equations. We define $\mathbf{R}^{(j)}(\mathbf{U}_i^n, \mathbf{Q}^n)$ as the solution at intermediate stage j at time $t = t_n + c_k \Delta t$ for the general r -order r -stages explicit RK method applied to (7), where $j, k = 1, 2, \dots, r$. Here, $\mathbf{U}_i^n$ represents the solution $\mathbf{U}(\mathbf{x}_i, t_n)$ and $\mathbf{Q}^n = \{\mathbf{U}_j^n, \mathbf{x}_j \in \mathcal{H}_{\mathbf{x}_i}\}$. The following equations are used for the RK method:

$$\begin{aligned} \mathbf{R}^{(j)}(\mathbf{U}_i^n, \mathbf{Q}^n) &= \mathbf{U}_i^n + \Delta t \sum_{k=1}^r a_{jk} \mathcal{L}_i(\mathbf{R}^{(k-1)}, t_n + c_k \Delta t), \quad j = 1, \dots, r, \\ \mathbf{U}_i^{n+1} &= \mathbf{U}_i^n + \Delta t \sum_{k=1}^r b_k \mathcal{L}_i(\mathbf{R}^{(k)}, t_n + c_k \Delta t), \end{aligned} \quad (10)$$

where $\mathbf{R}^{(j)} = \{\mathbf{R}^{(j)}(\mathbf{U}_i^n, \mathbf{Q}^n), \mathbf{x}_j \in \mathcal{H}_{\mathbf{x}_i}\}$, and constants c_j, a_{ij} and b_k satisfy

$$c_j = \sum_{k=1}^r a_{jk}, \quad \sum_{k=1}^r b_k = 1, \quad j = 1, \dots, r. \quad (11)$$

The RK method can be represented compactly in the Butcher Tableau, as shown in Table 1.

A thorough description and analysis of the coefficients are given in [46]. In this case, we opt for a one-parameter family of a third-order, three-stages ($r = 3$) explicit Runge–Kutta (RK) method, referred to as RK3, for the articulation of (1), as illustrated in Table 2. To enhance the accuracy and stability of the solutions, an explicit fourth-order, four-stages ($r = 4$) Runge–Kutta method, referred to as RK4, has also been adopted, as demonstrated in Table 3.

High-order RK methods in PD temporal discretization offer superior accuracy and lower computational times than other approaches. However, the PD model often requires a finer time step size to handle crack-related issues, resulting in increased

Table 3
Butcher Tableau for RK4.

0				
1/2	1/2			
1/2	0	1/2		
1	0	0	1	
	1/6	1/3	1/3	1/6

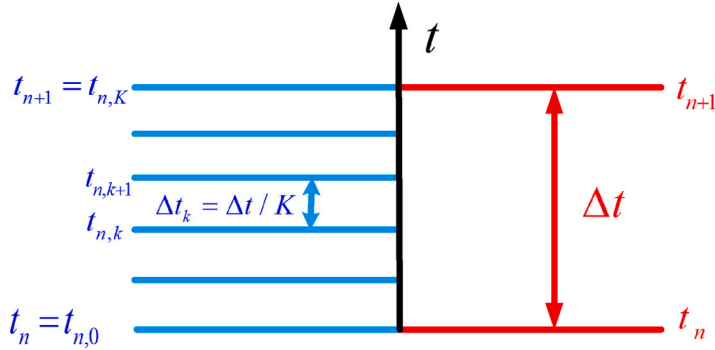


Fig. 1. Different time increments for the (fine local) sub-time step size and the (coarse global) time step size Δt .

computational load. Striking a balance between accuracy and computational efficiency is key in PD simulations. A multi-time-step scheme (MTS) rooted in the RK method effectively solves this challenge, adapting fine-time step sizes around cracks to maintain accuracy while reducing computational costs.

3. Multi-time-step schemes

This section presents the high-order multi-time-step (MTS) scheme applied to a rectangular Cartesian domain denoted as Ω . The domain Ω is divided into two subdomains: $\hat{\Omega}_C$ and $\hat{\Omega}_F$, that is, $\Omega = \hat{\Omega}_C \cup \hat{\Omega}_F$. Within $\hat{\Omega}_C$, a (coarse global) time step size Δt is assigned, while within $\hat{\Omega}_F$, a (fine local) sub-time step size Δt_k is specified for $k = 0, \dots, K$, where $K \geq 1$ is the refinement level. For $K = 1$, no refinement is made and the MTS scheme becomes the STS scheme. Generally, the MTS scheme becomes effective for $K \geq 2$ even for problem with discontinuous solution (crack) as shown in Section 4.3. We also note that the relationship $\Delta t = K \Delta t_k$ is enforced with $t_{n,k} = t_n + k \Delta t_k$, $k = 0, \dots, K$ (e.g., $t_n = t_{n,0}$ and $t_{n+1} = t_{n,K} = t_n + \Delta t$).

To address the nonlocal nature of the PD model and potential interactions between subdomains with different time step sizes, we further divide the entire domain Ω into four subdomains ($\Omega_F, \Omega_{FI}, \Omega_{CI}, \Omega_C$) as depicted in Fig. 2. Subdomains Ω_{FI} and Ω_{CI} act as internal boundary layers (BL).

- $\Omega_F = \left\{ \mathbf{x}_i : \mathbf{x}_i \in \hat{\Omega}_F, \left(\hat{\Omega}_F \cap H_{\mathbf{x}_i} \right) = H_{\mathbf{x}_i} \right\}$;
- $\Omega_{FI} = \left\{ \mathbf{x}_i : \mathbf{x}_i \in \hat{\Omega}_F, \left(\hat{\Omega}_F \cap H_{\mathbf{x}_i} \right) \neq H_{\mathbf{x}_i} \right\}$;
- $\Omega_{CI} = \left\{ \mathbf{x}_i : \mathbf{x}_i \in \hat{\Omega}_C, \left(\hat{\Omega}_C \cap H_{\mathbf{x}_i} \right) \neq H_{\mathbf{x}_i} \right\}$;
- $\Omega_C = \left\{ \mathbf{x}_i : \mathbf{x}_i \in \hat{\Omega}_C, \left(\hat{\Omega}_C \cap H_{\mathbf{x}_i} \right) = H_{\mathbf{x}_i} \right\}$.

As outlined below, we shall use different numerical methods to compute material points in different subdomains based on their distinct properties.

- Material points in subdomains $\Omega_C \cup \Omega_F$, which are not affected by subdomain $\hat{\Omega}_F$ with a sub-time step size Δt_k , are computed using the RK method with their respective time step size, that is, Δt for Ω_C and Δt_k for Ω_F .
- Material points in BL subdomain Ω_{CI} have a time step size of Δt but are influenced by subdomain $\hat{\Omega}_F$ that has a sub-time step size Δt_k . For this subdomain, we incorporate *correction term* as shown in (13).
- Material points in BL subdomain Ω_{FI} have a sub-time step size of Δt_k and are affected by subdomain $\hat{\Omega}_C$ that has a time step size Δt . However, since the solution advances from $t_{n,k}$ to $t_{n,k+1}$ with a sub-time step size Δt_k , the intermediate solutions in Ω_{CI} are unavailable. To compensate for this, we use *high-order interpolation polynomials* to estimate these missing intermediate solutions as depicted in Fig. 1.

Therefore, integrating methods from different subdomains is essential for constructing the MTS scheme. In the following sections, we will provide specific advanced stages for third- and fourth-order MTS schemes and prove their accuracy.

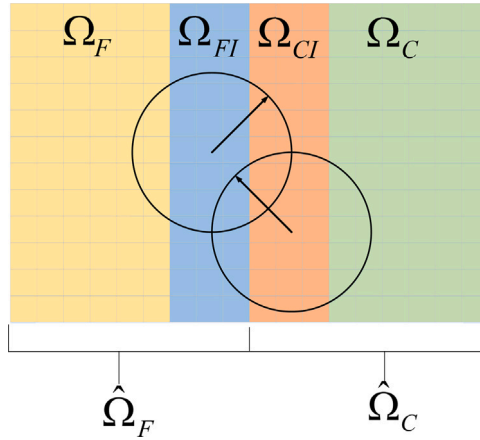


Fig. 2. Four subdomains of the physical domain Ω under uniform grid with $\delta = 3\Delta x$. The subdomain Ω_F , BL subdomain Ω_{FI} , BL subdomain Ω_{CI} , and subdomain Ω_C are colored in yellow, blue, red, and green, respectively. Here $\hat{\Omega}_C = \Omega_C \cup \Omega_{CI}$ and $\hat{\Omega}_F = \Omega_F \cup \Omega_{FI}$.

3.1. The third- and fourth-order multi-time-step (MTS) scheme

In this section, we introduce the third-order and fourth-order MTS schemes as solutions to address the challenge of varying time step sizes. To begin the MTS scheme, the solutions at t_0 and t_1 are defined as follows:

$$\mathbf{U}_i^0 = (\mathbf{u}^0, \mathbf{v}^0)^T, \quad \mathbf{U}_i^1 = \mathbf{U}_i^0 + \Delta t \sum_{k=1}^r b_k \mathcal{L}_i(\mathbf{R}^{(j)}, t_n + c_k \Delta t), \quad (12)$$

where $\mathbf{u}^0$ and $\mathbf{v}^0$ is the displacement and velocity at t_0 , respectively. $\mathbf{R}^{(j)}$ is obtained by the (10) when $n = 0$. Then the MTS scheme consists of three steps: first, advance $\mathbf{U}_i^n$ from t_n to t_{n+1} ($n \geq 2$) for the material points on $\hat{\Omega}_C$, next utilize their solutions to construct an interpolation polynomial and update the intermediate solution at the fine time step sizes, and finally advance on the subdomain $\hat{\Omega}_F$.

(1) Advancing from t_n to t_{n+1} ($n \geq 2$) on the subdomain $\hat{\Omega}_C$

Due to the influence of material points on the BL subdomain Ω_{FI} , the advancement from t_n to t_{n+1} of material points in the BL subdomain Ω_{CI} requires the inclusion of a *correction term* in the expression for $\mathbf{R}_i^{(j)}(\mathbf{U}_i^n, \mathbf{Q}^n)$, where $j = 1, \dots, r$. Let us denote $\bar{\Omega}_C = \hat{\Omega}_C \cup \Omega_{FI}$, $\mathbf{Q}_C^i = \mathbf{U}^i|_{\bar{\Omega}_C}$, and $\mathbf{Q}_C^{(j)} = \mathbf{U}^{(j)}|_{\bar{\Omega}_C}$. For a r -order r -stages RK method on the subdomain $\bar{\Omega}_C$, the following expression can be obtained for the value at the intermediate stages $\mathbf{U}_i^{(j)}$:

$$\mathbf{U}_i^{(j)} = \begin{cases} \mathbf{R}_i^{(j)}(\mathbf{U}_i^n, \mathbf{Q}_C^n), & \mathbf{x}_i \in \Omega_{CI}, \\ \mathbf{U}_i^n + \Delta t \sum_{k=1}^r a_{jk} \mathcal{L}_i(\mathbf{D}_C^{(k-1)}, t_n + c_k \Delta t), & \end{cases} \quad (13)$$

where $\mathbf{D}_C^{(j)} = \{\mathbf{R}^{(j)}(\mathcal{L}_i(\mathbf{Q}_C^n), \mathbf{G}^n), \mathbf{x}_j \in \mathcal{H}_{\mathbf{x}_i}\}$, $\mathbf{G}_C^n = \{\mathcal{L}_j(\mathbf{Q}_C^n), \mathbf{x}_j \in \mathcal{H}_{\mathbf{x}_i}\}$. For the derivatives in $\mathbf{D}_C^{(j)}$, we use the backward difference scheme to estimate, as follows:

$$\begin{aligned} \mathcal{L}''(\mathbf{Q}_C^n) &= \frac{\mathcal{L}_i(\mathbf{Q}_C^n) - \mathcal{L}_i(\mathbf{Q}_C^{n-1})}{\Delta t}, \\ \mathcal{L}'''(\mathbf{Q}_C^n) &= \frac{\mathcal{L}_i(\mathbf{Q}_C^n) - \mathcal{L}_i(\mathbf{Q}_C^{n-1})}{\Delta t^2} - \frac{\mathcal{L}_i(\mathbf{Q}_C^{n-1}) - \mathcal{L}_i(\mathbf{Q}_C^{n-2})}{\Delta t^2}, \end{aligned} \quad (14)$$

then the value $\mathbf{U}_i^{n+1}$ for $\mathbf{x}_i \in \hat{\Omega}_C$ can be obtained by

$$\mathbf{U}_i^{n+1} = \mathbf{U}_i^n + \begin{cases} \frac{\Delta t}{4} \left(\mathcal{L}_i(\mathbf{Q}_C^n) + \mathcal{L}_i(\mathbf{Q}_C^{(1)}) + \mathcal{L}_i(\mathbf{Q}_C^{(2)}) \right), & r=3, \\ \frac{\Delta t}{6} \left(\mathcal{L}_i(\mathbf{Q}_C^n) + 2\mathcal{L}_i(\mathbf{Q}_C^{(1)}) + 2\mathcal{L}_i(\mathbf{Q}_C^{(2)}) + \mathcal{L}_i(\mathbf{Q}_C^{(3)}) \right), & r=4. \end{cases} \quad (15)$$

Therefore, the value $\mathbf{U}_i^n$ is advanced from t_n to t_{n+1} on the subdomain $\hat{\Omega}_C$.

(2) Update the intermediate solution $\mathbf{U}_i^{n,k}$ on the BL subdomain Ω_{CI}

To make progress within the subdomain $\hat{\Omega}_F$, the computation of intermediate temporal values $\mathbf{U}_i^{n,k}$ for $\mathbf{x}_i \in \Omega_{IC}$ is necessary. Here, the index k ranges from 1 to K , representing the number of intermediate time levels. To predict $\mathbf{U}_i^{n,k}$, an interpolation technique is employed using an interpolating polynomial $I_i(t)$. It is required that this polynomial satisfies specific properties

for a r -order r -stages Runge–Kutta explicit scheme, as follows.

$$\mathcal{I}_i(t_{n,k}) = \mathbf{U}_i^{n,k} + O(\Delta t^{r+1}). \quad (16)$$

In this context, it is assumed that the operator $\mathcal{L}$ is sufficiently smooth, which allows us to employ the Taylor expansion for constructing the interpolation polynomial:

$$\mathbf{U}_i^{n,k} = \mathbf{U}_i^n + \mathcal{L}_i(\mathbf{Q}_C^n)(t_{n,k} - t_n) + \cdots + \mathcal{L}_i^{(r-1)}(\mathbf{Q}_C^n) \frac{(t_{n,k} - t_n)^r}{r!} + O(\Delta t^{r+1}). \quad (17)$$

Let us define $\mathbf{A}^r$ as an $(r-1) \times (r-1)$ matrix, and $\mathbf{f}^r$ and $\mathbf{d}^r$ as $(r-1)$ -dimensional vectors, where $\mathbf{d}_i^r = \mathcal{L}^{(i-1)}(\mathbf{Q}_C^n)$. Then, the derivative of $\mathcal{L}_i$ can be estimated as:

$$\mathbf{f}^r = \mathbf{A}^r \mathbf{d}^r + O(\Delta t^{r+1}). \quad (18)$$

We remark that the entries f_i^r in the vector $\mathbf{f}^r$ and a_{ij}^r in the matrix $\mathbf{A}^r$ can be obtained easily using the Taylor series expansion. Specifically, for the fourth-order scheme ($r = 4$), one has

$$\mathbf{f}^4 = \begin{bmatrix} \frac{\mathbf{U}_i^{n+1} - \mathbf{U}_i^n - \Delta t \mathcal{L}_i(\mathbf{Q}_C^n)}{\Delta t^2} \\ \frac{\mathcal{L}_i(\mathbf{Q}_C^n) - \mathcal{L}_i(\mathbf{Q}_C^{n-1})}{\Delta t} \\ \frac{\mathcal{L}_i(\mathbf{Q}_C^{n-1}) - \mathcal{L}_i(\mathbf{Q}_C^{n-2})}{\Delta t} \end{bmatrix}, \quad \mathbf{A}^4 = \begin{bmatrix} \frac{1}{2} & \frac{\Delta t}{6} & \frac{\Delta t^2}{24} \\ 1 & -\frac{\Delta t}{2} & \frac{\Delta t^2}{6} \\ 1 & -\frac{3\Delta t}{2} & \frac{7\Delta t^2}{6} \end{bmatrix}, \quad (19)$$

then $\mathbf{d}^4$ can be obtain by $\mathbf{d}^r = (\mathbf{A}^r)^{-1} \mathbf{f}^r + O(\Delta t^{r+1})$, where

$$(\mathbf{A}^4)^{-1} = \begin{bmatrix} \frac{8}{9} & \frac{37}{54} & -\frac{7}{54} \\ \frac{8}{3\Delta t} & -\frac{13}{9\Delta t} & \frac{1}{9\Delta t} \\ \frac{8}{3\Delta t^2} & -\frac{22}{9\Delta t^2} & \frac{10}{9\Delta t^2} \end{bmatrix}. \quad (20)$$

Substituting results into (17), an interpolation polynomial can be obtained as follows when $t_n \leq t \leq t_{n+1}$:

$$\mathcal{I}_i(t) = \mathbf{U}_i^n + (t - t_n) \mathcal{L}_i(\mathbf{Q}_C^n) + \sum_{j=2}^{r-1} \frac{(t - t_n)^j}{j!} \beta_{j-1}, \quad (21)$$

with

$$\beta_j = \sum_{i=1}^{r-1} \gamma_{ji} f_i, \quad (22)$$

where γ_{ij} is the entry in $(\mathbf{A}^r)^{-1}$. Finally, one has $\mathbf{U}_i^{n,k} = \mathcal{I}_i(t_{n,k})$, $\mathbf{x}_i \in \Omega_{CI}$.

Remark 1. For the third-order ($r = 3$) scheme, one set

$$a_{ij}^3 = a_{ij}^4, \quad f_i^3 = f_i^4. \quad (23)$$

then $\mathbf{d}^3$ can be obtained in the same manner as $\mathbf{d}^4$, where

$$(\mathbf{A}^3)^{-1} = \begin{bmatrix} \frac{6}{5} & \frac{2}{5} \\ \frac{12}{5\Delta t} & -\frac{6}{5\Delta t} \end{bmatrix}, \quad (24)$$

and $\mathcal{I}_i(t)$ is computed by (21).

(3) Advancing from t_n to t_{n+1} on the subdomain $\hat{\Omega}_F$

By constructing $\mathbf{U}^{n,k}$ through interpolation polynomials, the expression $\mathbf{R}_i^{(j)}(\mathbf{U}_i^n, \mathbf{Q}^n)$, where $j = 1, \dots, r$ and $\mathbf{x}_i \in \Omega_{IC}$, can be rewritten to advance the material points on $\hat{\Omega}_F$. Let $\bar{\Omega}_F = \Omega_{CI} \cup \hat{\Omega}_F$, $\mathbf{Q}_F^i = \mathbf{U}^i|_{\bar{\Omega}_F}$, and $\mathbf{Q}_F^{(i)} = \mathbf{U}^{(i)}|_{\bar{\Omega}_F}$. The solution at the intermediate stages is then computed using the Runge–Kutta method for $k = 1, 2, \dots, K$ ($K > 1$ is the refinement level):

$$\bar{\mathbf{U}}_i^{(j,k)} = \begin{cases} \mathbf{R}_i^{(j)}(\mathbf{U}_i^{n,k}, \mathbf{Q}_F^{n,k}), & \mathbf{x}_i \in \hat{\Omega}_F, \\ \mathbf{U}_i^n + \Delta t \sum_{k=1}^r a_{jk} \mathcal{L}_i(\mathbf{D}_F^{(k-1)}, t_n + c_k \Delta t), & \mathbf{x}_i \in \Omega_{CI}, \end{cases} \quad (25)$$

where $\mathbf{R}_i^{(j)}$ is obtained by Tables 2 and 3 for RK3 and RK4, respectively, and where $\mathbf{D}_F^{(j)} = \{\mathbf{R}_i^{(j)}(\mathbf{I}_i(t_{n,k}), \mathbf{I}), \mathbf{x}_j \in \mathcal{H}_{\mathbf{x}_i}\}$, $\mathbf{I} = \{\mathbf{I}_j(t_{n,k}), \mathbf{x}_j \in \mathcal{H}_{\mathbf{x}_i}\}$. Then the value $\mathbf{U}_i^n$ can be advanced from t_n to t_{n+1} by

$$\mathbf{U}_i^{n,k+1} = \mathbf{U}_i^{n,k} + \begin{cases} \frac{\Delta t_k}{4} \left(\mathcal{L}_i(\mathbf{Q}_F^{n,k}) + \frac{3}{2} \mathcal{L}_i(\mathbf{Q}_F^{(1,k)}) + \frac{3}{2} \mathcal{L}_i(\mathbf{Q}_F^{(2,k)}) \right), & r=3, \\ \frac{\Delta t_k}{6} \left(\mathcal{L}_i(\mathbf{Q}_F^{n,k}) + 2\mathcal{L}_i(\mathbf{Q}_F^{(1,k)}) + 2\mathcal{L}_i(\mathbf{Q}_F^{(2,k)}) + \mathcal{L}_i(\mathbf{Q}_F^{(3,k)}) \right), & r=4, \end{cases} \quad (26)$$

as a result of the aforementioned computations, the value $\mathbf{U}_i^{n+1}$ is obtained for all material points $\mathbf{x}_i \in \Omega$.

3.2. Convergence analysis

The STS and MTS schemes exhibit the same convergence behavior in smooth cases. The third- and fourth-order MTS follow the theorem below:

Theorem 1. *If the operator $\mathcal{L}$ is C^5 , the global error of the r -order MTS under the uniform time step size Δt is $O(\Delta t^r)$, where $\mathbf{U}(\mathbf{x}, t)$ is the exact solution of the semi-discrete system in (7). It holds for $r = 3, 4$.*

Proof. Since the third-order scheme is a special form of the fourth-order scheme, we first provide proof of the fourth-order scheme. Without loss of generality, we assume $\mathbf{e}_i^m = 0$ and $\mathbf{e}_i^{m,n} = 0$ for $m = 1, 2, \dots, n$ and $n = 1, 2, \dots, k$.

- (i) First, we prove that the local error for the material points on the subdomain $\hat{\Omega}_C$ is $O(\Delta t^5)$. Based on the construction process of the MTS scheme, the following expression is obtained:

$$\begin{aligned} \mathbf{U}_i^{n+1} &= \mathbf{U}_i^{n+1} + \frac{\Delta t}{6} \left(\mathcal{L}_i(\mathbf{Q}_C^n) + 2\mathcal{L}_i(\mathbf{Q}_C^{(1)}) + 2\mathcal{L}_i(\mathbf{Q}_C^{(2)}) + \mathcal{L}_i(\mathbf{Q}_C^{(3)}) \right) \\ &= \mathbf{U}_i^{n+1} + \Delta t \mathcal{L}(\mathbf{Q}_C^n) + \frac{\Delta t^2}{2} \mathcal{L}'(\mathbf{Q}_C^n) + \frac{\Delta t^3}{3!} \mathcal{L}''(\mathbf{Q}_C^n) + \frac{\Delta t^4}{4!} \mathcal{L}'''(\mathbf{Q}_C^n) + O(\Delta t^5) \\ &= \hat{\mathbf{U}}_i^n + \Delta t \hat{\mathbf{U}}_i'(t_n) + \frac{\Delta t^2}{2} \hat{\mathbf{U}}_i''(t_n) + \frac{\Delta t^3}{3!} \hat{\mathbf{U}}_i'''(t_n) + \frac{\Delta t^4}{4!} \hat{\mathbf{U}}_i''''(t_n) + O(\Delta t^5) \\ &= \hat{\mathbf{U}}_i(t_{n+1}) + O(\Delta t^5), \quad \mathbf{x}_i \in \Omega_{CI}. \end{aligned} \quad (27)$$

Thus the local error $\mathbf{e}_i^{n+1}$ for the material points on the BL subdomain Ω_{CI} is $O(\Delta t^5)$. It can be observed that the value of the material points on the subdomain Ω_C is obtained by RK4; thus, the local error is also $O(\Delta t^5)$.

- (ii) With the error obtained by the interpolation polynomial, the local error on the subdomain $\hat{\Omega}_F$ can be demonstrated as follows:

$$\mathbf{I}_i(t_{n,k}) = \hat{\mathbf{U}}_i(t_{n,k}) + O(\Delta t_k^5). \quad (28)$$

Substituting into (25), we have

$$\bar{\mathbf{U}}_i^{(j,k)} = \begin{cases} \mathbf{R}_i^{(j)}(\mathbf{U}_i^{n,k}, \mathbf{Q}_F^{n,k}), & \mathbf{x}_i \in \hat{\Omega}_F, \\ \mathbf{R}_i^{(j)}(\mathbf{U}_i^{n,k}, \hat{\mathbf{Q}}'(t_{n,k})) + O(\Delta t_k^5), & \mathbf{x}_i \in \Omega_{CI}, \end{cases} \quad (29)$$

where $\hat{\mathbf{Q}}'(t_{n,k}) = \{\hat{\mathbf{U}}_i'(t_{n,k}), \mathbf{x}_j \in \mathcal{H}_{\mathbf{x}_i}\}$. By Taylor expansion, we can obtain follows

$$\mathbf{U}_i^{n,k+1} = \hat{\mathbf{U}}_i(t_{n,k+1}) + O(\Delta t_k^5), \quad \mathbf{x}_i \in \hat{\Omega}_F. \quad (30)$$

Then, the local error of the material points on the subdomain $\bar{\Omega}_F$ is $O(\Delta t_k^5) = O(\Delta t^5)$.

- (iii) The global error of the material points in the domain Ω can be obtained by considering the local error. Let $\mathbf{E}^{n+1}$ represent the value at t_{n+1} of the local error committed at $t_{n,k}$ and propagated forward by the numerical scheme. The local error $\mathbf{e}_i^{n+1}$ can be expressed as the sum of $\mathbf{E}_i^{n,k}$ for $k = 1, 2, \dots, K$. Thus, the accumulated error can be estimated as follows:

$$|\mathbf{e}_i^{n+1}| \leq \sum_{k=1}^K |\mathbf{E}_i^{n,k}|. \quad (31)$$

According to Theorem I.10.2 in [47], it can be shown that:

$$|\mathbf{E}_i^{n,k}| \leq \mathbf{e}_i^{B(t_{n+1}-t_{n,k})} |\mathbf{e}_i^{n,k}| \leq \mathbf{e}_i^{B\Delta t} |\mathbf{e}_i^{n,k}|, \quad (32)$$

where B is an upper bound on the Jacobian of (7). Since $|\mathbf{e}_i^{n,k}| \leq C_2 \Delta t^4$, the estimate becomes:

$$|\mathbf{e}_i^{n+1}| \leq K \mathbf{e}_i^{B\Delta t} |\mathbf{e}_i^{n,k}| \leq K \left(1 + \Delta t B + \frac{(\Delta t B)^2}{2} + \dots \right) C_2 \Delta t^5 \leq C_3 \Delta t^5, \quad (33)$$

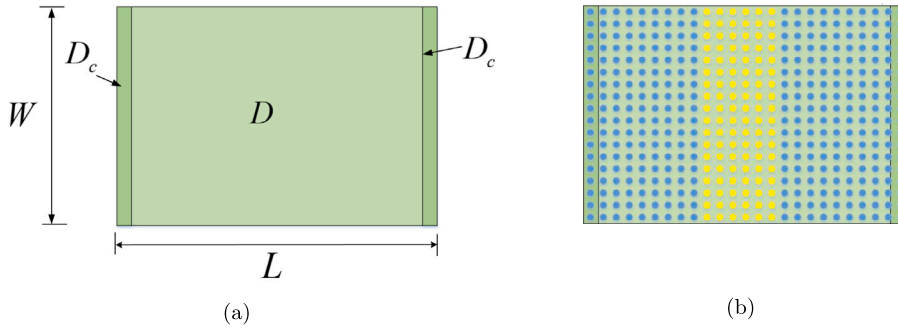


Fig. 3. Geometry of a 2D plate under transverse loading and its discretization.

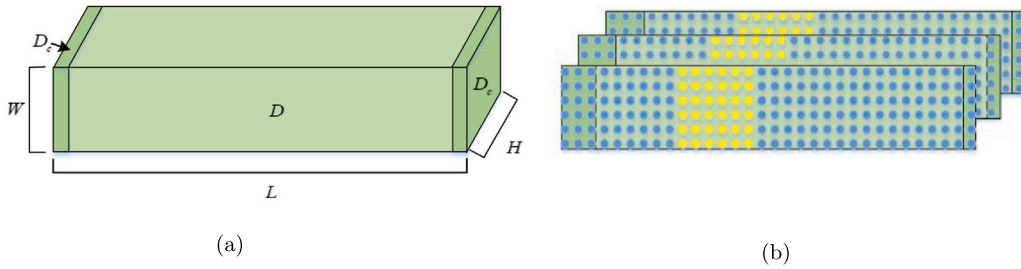


Fig. 4. Geometry of a 3D plate under transverse loading and its discretization.

here C_2 and C_3 are some constants. Combining this with Theorem II.3.4 in [47], we obtain the expression for the global error $\mathbf{E}_i$:

$$|\mathbf{E}_i| \leq \frac{C_2}{K} (\mathbf{e}^{BT} - 1) \Delta t^4. \quad (34)$$

Similarly, one can also show that the global error of the third-order MTS is $O(\Delta t^3)$. Thus, the theorem is proved. $\square$

4. Numerical results

In this section, we present three examples to demonstrate the accuracy and convergence of the MTS scheme for bond-based PD on various geometries. The numerical experiments below are implemented in Fortran[®] and conducted on a workstation with an Intel Xeon Gold 6240 CPU operating at 2.6 GHz and 2048 GB of installed memory.

4.1. Plate under transverse loading

In this section, we will consider a 2D linear bond-based PD applied to a plate under uniaxial tension. The plate has dimensions of $W = 1$ m (width), $L = 0.5$ m (length), and $d = 0.01$ m (thickness). The geometry of the plate is illustrated in Fig. 3.

The material properties are $E = 1.92 \times 10^{11}$ Mpa (elastic modulus), $\nu = 1/3$ (Poisson's ratio) and $\rho = 8000$ kg/m³ (density). The external loading $b_p = p_0 W/d$ is applied to boundary layer D_c , where the uniaxial tension loading $p_0 = 200$ Mpa, and the width of D_c is d . The horizon size δ is chosen as 0.03 m.

For the spatial discretization of the model, a mesh with a size of 100×50 is used. The yellow subdomain in Fig. 3(b) is computed using the MTS scheme with sub-time step size $\Delta t_k, k = 1, \dots, K$ where K is the refinement level, while the displacements in the other areas are obtained using the time step size Δt . The L^2 absolute error with the exact solution obtained using the displacement in the x -direction from the STS scheme with a fine time step size $\Delta t_e = 8 \times 10^{-3} \times \sqrt{\frac{2\rho}{\pi\delta^2\alpha}}$ s. To compare the performance of the STS and MTS schemes, we refine the time step sizes Δt gradually from $100\Delta t_e$ to $12.5\Delta t_e$, and the total simulation time considered is $1 \times 10^5 \Delta t_e$.

Table 4 provides the error and order for the two-dimensional problem at different time step sizes. The comparison between the STS ($K = 1$) and MTS ($K > 1$) schemes reveals that the relative error decreases as the refinement level K increases. It should be noted that when $K = 1$, the proposed MTS scheme is the classical RK method. This indicates that refining Δt by increasing K leads to improved accuracy. For example, when refining K from two to eight with $\Delta t = 50\Delta t_e$ s, the error reduces from 8.87×10^{-11} to 5.02×10^{-11} , highlighting the accuracy of the MTS scheme.

Furthermore, an interesting observation is that the orders of the STS and MTS schemes are well preserved. It means that the order of the MTS scheme, whether based on RK3 or RK4, remains consistent even when the time step size Δt varies. This finding validates the efficacy of the analysis of numerical schemes and demonstrates that the MTS scheme maintains its expected order of accuracy across different time step sizes.

Table 4

The error and order (CR) obtained by MTS based on RK3 (MTS3) and RK4 (MTS4) with different Δt and different refinement level K on a 2D model.

Δt (s)	K	MTS3		MTS4	
		Error	CR	Error	CR
$100\Delta t_e$	1	3.28E-10	–	7.62E-11	–
	2	2.54E-10	–	4.45E-11	–
	4	2.41E-10	–	4.37E-11	–
	8	2.40E-10	–	4.35E-11	–
$50\Delta t_e$	1	8.87E-11	3.07	5.45E-12	3.91
	2	5.48E-11	3.07	3.16E-12	3.91
	4	5.07E-11	3.08	3.04E-12	3.93
	8	5.02E-11	3.09	3.03E-12	3.93
$25\Delta t_e$	1	1.28E-11	3.03	3.46E-13	3.96
	2	7.66E-12	3.04	1.99E-13	3.98
	4	7.10E-12	3.04	1.92E-13	3.97
	8	7.03E-12	3.04	1.91E-13	3.97
$12.5\Delta t_e$	1	1.63E-12	3.00	2.17E-14	3.98
	2	9.71E-13	3.01	1.25E-14	3.98
	4	8.99E-13	3.00	1.20E-14	3.99
	8	8.91E-13	3.00	1.20E-14	3.99

Table 5

The error and order (CR) obtained by the MTS scheme based on RK3 (MTS3) and RK4 (MTS4) with different Δt and different refinement level K on a 3D model.

Δt (s)	K	MTS3		MTS4	
		Error	CR	Error	CR
1×10^{-6}	1	8.70E-08	–	7.72E-09	–
	2	6.21E-08	–	5.48E-09	–
	4	6.15E-08	–	5.46E-09	–
	8	6.15E-08	–	5.46E-09	–
0.5×10^{-6}	1	1.14E-08	3.12	4.89E-10	3.99
	2	8.13E-09	3.13	3.47E-10	4.02
	4	8.06E-09	3.13	3.46E-10	4.01
	8	8.06E-09	3.13	3.46E-10	4.01
0.25×10^{-6}	1	1.43E-10	3.12	3.07E-11	4.00
	2	1.02E-10	3.03	2.17E-11	4.00
	4	1.01E-10	3.03	2.17E-11	4.00
	8	1.01E-10	3.03	2.17E-11	4.00
0.125×10^{-6}	1	1.78E-10	3.00	1.92E-12	3.99
	2	1.27E-10	3.01	1.36E-12	4.00
	4	1.26E-10	3.00	1.35E-13	3.99
	8	1.26E-10	3.00	1.35E-13	3.99

4.2. Block under transverse loading

In this section, we conducted a PD simulation of a 3D block subjected to quasi-static transverse loading. The block has dimensions of 1.0 m in length, 0.1 m in width, and 0.1 m in thickness, as shown in Fig. 4(a). The horizon size δ was chosen as 0.03 m. The external loading of $b_p = p_0 W/d$ was applied to the area D_c , where the thickness d of D_c is 0.01 m, and the value of p_0 is 200 MPa. The material properties used in the simulation are an elastic modulus E of 2.0×10^5 MPa and a Poisson's ratio ν of $1/4$.

A $100 \times 10 \times 10$ mesh is used to discrete this model. To investigate the convergence of the MTS scheme, we partition the PD model into two distinct subdomains. The first subdomain, represented by the yellow part in Fig. 4(b), corresponded to the area using sub-time step size Δt_k with a thickness of 2δ . The second subdomain was computed using Δt . To calculate the L_2 error, the displacement obtained from the STS scheme when $\Delta t = 1 \times 10^{-8}$ s was chosen as the reference exact solution. The time step size was refined from 1×10^{-6} to 0.125×10^{-6} to examine the accuracy of the scheme.

Table 5 demonstrates the accuracy and convergence of the three-dimensional PD model using the MTS scheme. One notable finding is that the order of the MTS scheme based on the RK method remains consistent with the order of the STS scheme, regardless of the coarse and fine time step sizes. Furthermore, we observed that the displacement error decreases as the coarse-to-fine mesh size ratio increases. For example, when $\Delta t = 0.5 \times 10^{-6}$, the error decreases from 1.14×10^{-8} to 8.06×10^{-9} as the refinement level K is increased from 1 to 8. It highlights the accuracy benefits of employing the MTS scheme.

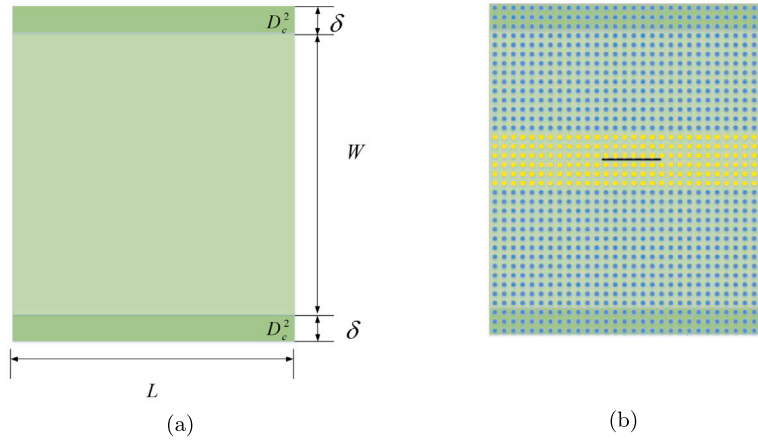


Fig. 5. Geometry of a 2D plate with a pre-existing crack (black line).

4.3. Plate with a pre-existing crack

In this example, we consider a 2D plate with a length of $L = 0.05$ m and a width of $W = 0.05$ m, which contains a pre-existing crack with a length of $2c = 0.01$ m, as shown in Fig. 5(a). The material properties, including density ρ , horizon size δ , elastic modulus E , and Poisson's ratio μ , are set to be consistent with Section 4.1. Velocity constraints are applied to D_c^1 and D_c^2 at the top and bottom of region D , respectively. These constraints have a depth of δ , as follows:

$$\begin{aligned} \dot{u}_y(\mathbf{x}_i, t) &= 20.0 \text{ m/s} & \mathbf{x}_i \in D_c^1, \\ \dot{u}_y(\mathbf{x}_i, t) &= -20.0 \text{ m/s} & \mathbf{x}_i \in D_c^2. \end{aligned} \quad (35)$$

For the spatial discretization of the model, a mesh of 500×500 is used. The history-dependent scalar function μ is used to identify fractures in the PD simulation. The function is multiplied by Eq. (1) to take into account the effects of previous deformation history on the material response and can be defined as follows:

$$\mu(s, t) = \begin{cases} 1, & \text{if the bond is not broken } s < s_0 \\ 0, & \text{if the bond is broken } s \geq s_0, \end{cases} \quad (36)$$

in which s_0 is the critical bond stretch and chosen as 0.04472, s is the bond stretches:

$$s = \frac{|\boldsymbol{\eta} + \boldsymbol{\xi}| - |\boldsymbol{\xi}|}{|\boldsymbol{\xi}|}. \quad (37)$$

In Fig. 6, the simulation outcomes are presented. The blue shades represent material points where the bonds remain intact despite the displacement changes, corresponding to $s = 1$. On the other hand, the yellow shades indicate regions where the bonds have been broken, resulting in $s = 0$. In certain areas where the material points have undergone significant stretching, the value of s is between 0 and 1, indicating partial bond breakage. These results demonstrate that the MTS scheme approach accurately captures the displacement changes and fracture behavior, similar to the STS scheme. The ability of the MTS scheme to handle complex deformation and fracture phenomena makes it a valuable tool in PD simulations.

The MTS scheme possesses a noteworthy advantage, namely its capability to attain computational accuracy on par with the solution by the STS scheme while reducing computational time. This advantageous characteristic is demonstrated in Fig. 7. For a selected number of time steps, specifically 1500 with a time increment of $\Delta t_e = 0.8\sqrt{(2\rho)/(\pi\delta^2\alpha)}$ s, the MTS scheme accomplishes the crack simulation in a mere 80 min of CPU time, compared to the 121 min required by the STS scheme. The time saved in computation is directly linked to the extent of the fine time step region. Notably, the MTS scheme exhibits enhanced computational efficiency as the fine time step region decreases in size.

Table 6 provides a summary of the absolute error, compared to the solution computed by the STS scheme, for the PD approach with a time step size of $\Delta t = 0.125\Delta t_e$. The table includes the regions' errors with coarse time step e_c , fine time step e_f , and both e . The comparison between the STS and MTS schemes for K from 2 to 8 is shown. As expected, we noted that decreasing the time step size leads to improved simulation accuracy.

The MTS scheme demonstrates high accuracy in both coarse and fine time step sizes, with improved accuracy observed in the $\hat{\Omega}_C$ subdomain when comparing $K = 1$ and $K = 8$. In the fine time step region $\hat{\Omega}_F$, the MTS scheme outperforms the STS scheme. Additionally, the MTS scheme can significantly speed up computational processes (up to four times in some examples) compared to the STS scheme due to its flexible time step size divisions. Various examples have shown that the MTS scheme maintains high accuracy while efficiently solving problems of different sizes and time-step ratios. Hence, it is recommended for PD simulations, particularly for large-scale problems.

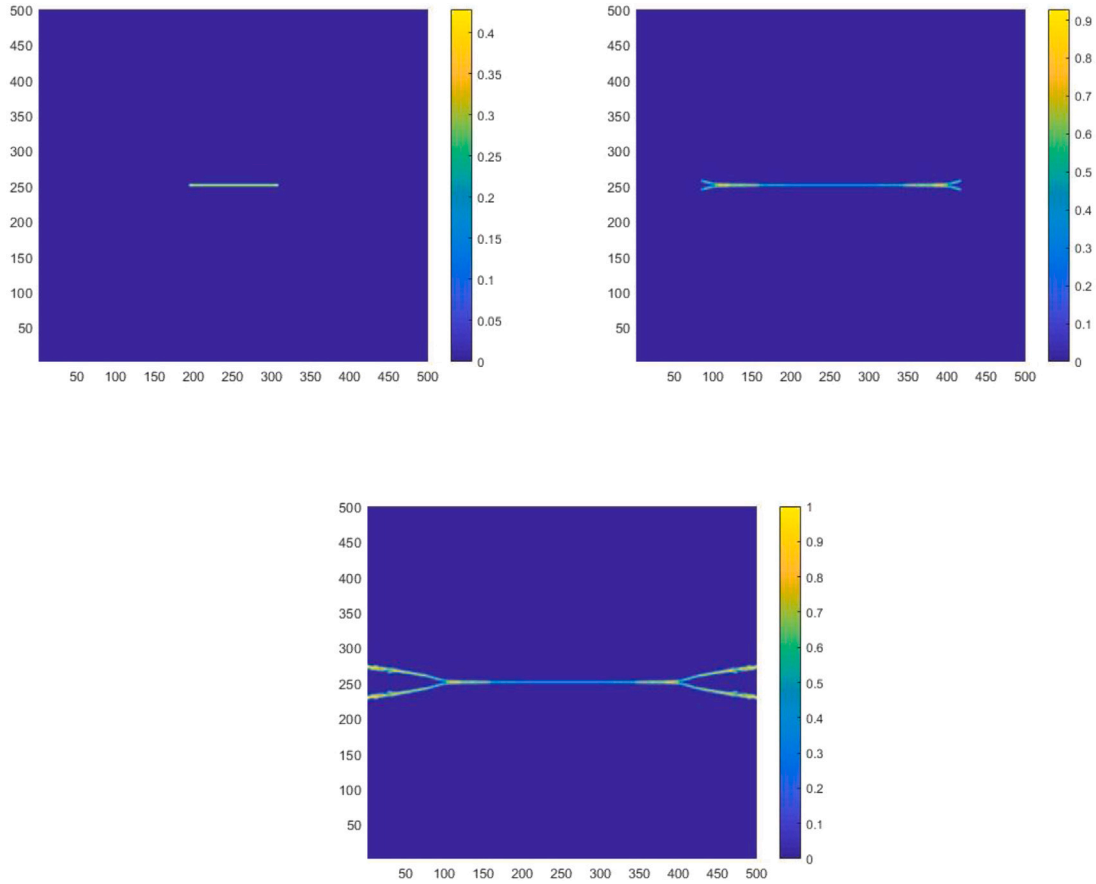


Fig. 6. Crack propagation simulation by MTS when the number of the time step is chosen as 500, 1000 and 1500 for $\Delta t = 0.5 \times 10^{-5}$ s.

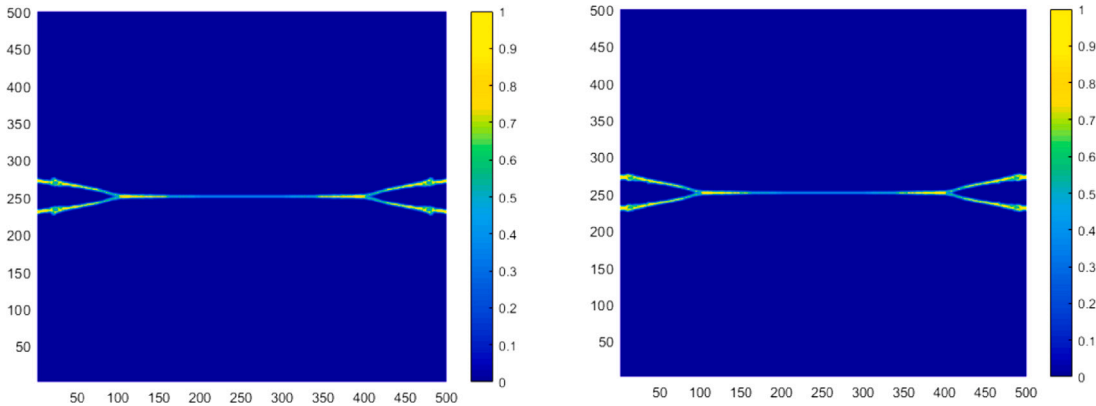


Fig. 7. Crack propagation simulation by MTS4 ($K = 2$ and CPU time = 80 m) and by RK4 (CPU time = 121 m) when the physical time is chosen as $1500\Delta t_c$.

5. Conclusions

This paper's main contribution is the introduction of a high-order MTS for peridynamics. This scheme addresses the challenge of achieving computational accuracy by utilizing smaller spatial grids and time steps in discontinuous regions and larger spatial grids and time steps in other areas. Thus, it overcomes the limitation of using a small single global time step for the entire simulation. Using a higher-order Runge–Kutta method further enhances the accuracy and efficiency of the MTS scheme. Theoretical analysis and numerical examples have validated the proposed method's stability, accuracy, and computational advantages. Through a Taylor expansion analysis, it has been demonstrated that the MTS scheme maintains the order of the higher-order Runge–Kutta method.

Table 6

The error of different subdomain obtained by the MTS scheme based on RK4 with different Δt , the number of time steps nt , and different refinement level K for a 2D crack model; e_c is the errors for the coarse time step region, e_f is the error for the fine time step region, and e is the error for the overall region.

Δt (s)	K		nt		
			500	1000	1500
Δt_e	8	e_f	5.43E-9	2.28E-7	2.88E-6
		e_c	8.68E-9	1.09E-7	4.91E-7
		e	1.03E-8	2.56E-7	2.92E-6
$0.5\Delta t_e$	4	e_f	2.57E-9	1.38E-7	1.30E-6
		e_c	3.72E-9	5.05E-8	1.81E-7
		e	4.56E-8	1.48E-7	3.32E-6
$0.25\Delta t_e$	2	e_f	1.07E-9	8.13E-8	7.40E-7
		e_c	1.24E-9	2.20E-8	1.49E-7
		e	1.65E-9	8.46E-8	7.56E-7

Several examples in two-dimensional and three-dimensional settings have been employed to verify that the STS and MTS schemes exhibit consistent convergence rates and achieve high accuracy at different time steps.

Data availability

Data will be made available on request.

Acknowledgments

H. Tian was supported by the Fundamental Research Funds for the Central Universities (202042008,202264006) and the National Natural Science Foundation of China (11971482). W.-S. Don would like to acknowledge the Shandong Provincial Natural Science Foundation (ZR2022MA012) and a grant from the Hong Kong Research Grant Council GRF for supporting this research.

References

- [1] S. Tang, H. Zhang, C. Tang, H. Liu, Numerical model for the cracking behavior of heterogeneous brittle solids subjected to thermal shock, *Int. J. Solids Struct.* 80 (2016) 520–531.
- [2] F. Meng, M. Thouless, Cohesive-zone analyses with stochastic effects, illustrated by an example of kinetic crack growth, *J. Mech. Phys. Solids* 132 (2019) 103686.
- [3] Y. Wang, Q. Wang, J.X. Wen, J. Sun, K. Liew, Investigation of thermal breakage and heat transfer in single, insulated and laminated glazing under fire conditions, *Appl. Therm. Eng.* 125 (2017) 662–672.
- [4] J.F. Unger, S. Eckardt, C. Könke, Modelling of cohesive crack growth in concrete structures with the extended finite element method, *Comput. Methods Appl. Mech. Engrg.* 196 (41) (2007) 4087–4100.
- [5] G. Zi, T. Rabczuk, W. Wall, Extended meshfree methods without branch enrichment for cohesive cracks, *Comput. Mech.* 40 (2007) 367–382.
- [6] L. Xia, D. Da, J. Yvonnnet, Topology optimization for maximizing the fracture resistance of quasi-brittle composites, *Comput. Methods Appl. Mech. Engrg.* 332 (2018) 234–254.
- [7] S. Bhowmick, G. Liu, Three dimensional CS-FEM phase-field modeling technique for brittle fracture in elastic solids, *Appl. Sci.* 8 (2018) 2488, <http://dx.doi.org/10.3390/app8122488>.
- [8] A. Egger, U. Pillai, K. Agathos, E. Kakouris, E. Chatzi, I. Ashcroft, S. Triantafyllou, Discrete and phase field methods for linear elastic fracture mechanics: A comparative study and state-of-the-art review, *Appl. Sci.* 9 (2019) 2436.
- [9] J. Xu, A. Askari, O. Weckner, S. Silling, Peridynamic analysis of impact damage in composite laminates, *J. Aerosp. Eng.* 21 (3) (2008) 187–194.
- [10] S. Silling, M. Epton, O. Weckner, J. Xu, E. Askari, Peridynamic states and constitutive modeling, *J. Elasticity* 88 (2007) 151–184.
- [11] G. Ongaro, R. Bertani, U. Galvanetto, A. Pontefisso, M. Zaccariotto, A multiscale peridynamic framework for modelling mechanical properties of polymer-based nanocomposites, *Eng. Fract. Mech.* 274 (2022) 108751.
- [12] Y. Hu, N. De Carvalho, E. Madenci, Peridynamic modeling of delamination growth in composite laminates, *Compos. Struct.* 132 (2015) 610–620.
- [13] V.A. Buryachenko, Generalized effective fields method in peridynamic micromechanics of random structure composites, *Int. J. Solids Struct.* 202 (2020) 765–786.
- [14] S. Jafarzadeh, Z. Chen, F. Bobaru, Computational modeling of pitting corrosion, *Corros. Rev.* 37 (5) (2019) 419–439.
- [15] Z. Chen, S. Jafarzadeh, J. Zhao, F. Bobaru, A coupled mechano-chemical peridynamic model for pit-to-crack transition in stress-corrosion cracking, *J. Mech. Phys. Solids* 146 (2021) 104203.
- [16] J. Zhao, S. Jafarzadeh, M. Rahmani, Z. Chen, Y.-R. Kim, F. Bobaru, A peridynamic model for galvanic corrosion and fracture, *Electrochim. Acta* 391 (2021) 138968.
- [17] S. Jafarzadeh, Z. Chen, S. Li, F. Bobaru, A peridynamic mechano-chemical damage model for stress-assisted corrosion, *Electrochim. Acta* 323 (2019) 134795.
- [18] S. Jafarzadeh, Z. Chen, J. Zhao, F. Bobaru, Pitting, lacy covers, and pit merger in stainless steel: 3D peridynamic models, *Corros. Sci.* 150 (2019) 17–31.
- [19] C.T. Nguyen, S. Oterkus, E. Oterkus, An energy-based peridynamic model for fatigue cracking, *Eng. Fract. Mech.* 241 (2021) 107373.
- [20] X. Li, X. Gu, X. Xia, E. Madenci, X. Chen, Q. Zhang, Effect of water-cement ratio and size on tensile damage in hardened cement paste: Insight from peridynamic simulations, *Constr. Build. Mater.* 356 (2022) 129256.
- [21] D. Jin, W. Liu, A peridynamic modeling approach of solid state impact bonding and simulation of interface morphologies, *Appl. Math. Model.* 92 (2021) 466–485.

- [22] M. Nowak, K. Mulewska, A. Azarov, A. Ustrzycka, et al., A peridynamic elasto-plastic damage model for ion-irradiated materials, *Int. J. Mech. Sci.* 237 (2023) 107806.
- [23] B. Ren, C. Wu, P. Seleson, D. Zeng, M. Nishi, M. Pasetto, An FEM-based peridynamic model for failure analysis of unidirectional fiber-reinforced laminates, *J. Peridyn. Nonlocal Model.* (2022) 1–20.
- [24] S.A. Silling, M. D'Elia, Y. Yu, H. You, M. Fermen-Coker, Peridynamic model for single-layer graphene obtained from coarse-grained bond forces, *J. Peridyn. Nonlocal Model.* (2022) 1–22.
- [25] S.A. Silling, M.L. Parks, J.R. Kamm, O. Weckner, M. Rassaian, Modeling shockwaves and impact phenomena with Eulerian peridynamics, *Int. J. Impact Eng.* 107 (2017) 47–57.
- [26] S.A. Silling, E. Askari, A meshfree method based on the peridynamic model of solid mechanics, *Comput. Struct.* 83 (17–18) (2005) 1526–1535.
- [27] S. Li, Y. Jin, X. Huang, L. Zhai, An extended bond-based peridynamic approach for analysis on fracture in brittle materials, *Math. Probl. Eng.* 2020 (2020) 1–12.
- [28] P. Diehl, S. Prudhomme, M. Lévesque, A review of benchmark experiments for the validation of peridynamics models, *J. Peridyn. Nonlocal Model.* 1 (2019) 14–35.
- [29] X. Tian, Q. Du, Analysis and comparison of different approximations to nonlocal diffusion and linear peridynamic equations, *SIAM J. Numer. Anal.* 51 (6) (2013) 3458–3482.
- [30] T. Bode, C. Weißenfels, P. Wriggers, Peridynamic Galerkin method: an attractive alternative to finite elements, *Comput. Mech.* 70 (4) (2022) 723–743.
- [31] J. Lu, Y. Nie, A reduced-order fast reproducing kernel collocation method for nonlocal models with inhomogeneous volume constraints, *Comput. Math. Appl.* 121 (2022) 52–61.
- [32] J. Lu, M. Yang, Y. Nie, Convergence analysis of Jacobi spectral collocation methods for weakly singular nonlocal diffusion equations with volume constraints, *Appl. Math. Comput.* 431 (2022) 127345.
- [33] S. Zhang, Y. Nie, Localized Chebyshev and MLS collocation methods for solving 2D steady state nonlocal diffusion and peridynamic equations, *Math. Comput. Simulation* 206 (2023) 264–285.
- [34] S. Jafarzadeh, F. Mousavi, A. Larios, F. Bobaru, A general and fast convolution-based method for peridynamics: applications to elasticity and brittle fracture, *Comput. Methods Appl. Mech. Engrg.* 392 (2022) 114666.
- [35] S. Jafarzadeh, L. Wang, A. Larios, F. Bobaru, A fast convolution-based method for peridynamic transient diffusion in arbitrary domains, *Comput. Methods Appl. Mech. Engrg.* 375 (2021) 113633.
- [36] M. Zaccariotto, T. Mudric, D. Tomasi, A. Shojaei, U. Galvanetto, Coupling of FEM meshes with peridynamic grids, *Comput. Methods Appl. Mech. Engrg.* 330 (2018) 471–497.
- [37] Y. Yu, F.F. Bargos, H. You, M.L. Parks, M.L. Bittencourt, G.E. Karniadakis, A partitioned coupling framework for peridynamics and classical theory: analysis and simulations, *Comput. Methods Appl. Mech. Engrg.* 340 (2018) 905–931.
- [38] J. Zhang, F. Han, Z. Yang, J. Cui, Coupling of an atomistic model and bond-based peridynamic model using an extended Arlequin framework, *Comput. Methods Appl. Mech. Engrg.* 403 (2023) 115663.
- [39] F. Han, G. Lubineau, Y. Azdoud, Adaptive coupling between damage mechanics and peridynamics: A route for objective simulation of material degradation up to complete failure, *J. Mech. Phys. Solids* 94 (2016) 453–472.
- [40] P. Seleson, Y.D. Ha, S. Beneddine, Concurrent coupling of bond-based peridynamics and the navier equation of classical elasticity by blending, *Int. J. Multiscale Comput. Eng.* 13 (2015) 91–113.
- [41] M. Birner, P. Diehl, R. Lipton, M.A. Schweitzer, A fracture multiscale model for peridynamic enrichment within the partition of unity method, *Adv. Eng. Softw.* 176 (2023) 103360.
- [42] C.J. Trahan, C. Dawson, Local time-stepping in Runge–Kutta discontinuous Galerkin finite element methods applied to the shallow-water equations, *Comput. Methods Appl. Mech. Engrg.* 217 (2012) 139–152.
- [43] P. Lindsay, M. Parks, A. Prakash, Enabling fast, stable and accurate peridynamic computations using multi-time-step integration, *Comput. Methods Appl. Mech. Engrg.* 306 (2016) 382–405.
- [44] H. Tian, J. Sun, C. Liu, X. Zheng, Accelerating convergence of crack propagation simulation in peridynamic models via high-order temporal discretization, *Appl. Math. Lett.* 145 (2023) 108759.
- [45] S. Silling, Reformulation of elasticity theory for discontinuities and long-range forces, *J. Mech. Phys. Solids* 48 (1) (2000) 175–209.
- [46] U.M. Ascher, L.R. Petzold, *Computer Methods for Ordinary Differential Equations and Differential-Algebraic Equations*, Society for Industrial and Applied Mathematics, Philadelphia, PA, 1998.
- [47] G.W.E. Hairer, S.P. Norsett, *Solving Ordinary Differential Equations I. Nonstiff Problems*, second ed., Springer, Berlin, 2000.